MEDGUIDE: Agentic Graph RAG for Evidence-Grounded and Explainable Clinical Decision Support

Dr. Madhuri Pagale, Vikram Kadam, Keshav Mittal, Sandesh Phad, and Tejas Shidam
Emails: {madhuri.pagale, vikram.kadam23, keshav.mittal23, sandesh.phad23, tejas.shidam23}@pccoepune.org

***Abstract*—While Large Language Models (LLMs) encode extensive biomedical knowledge, clinical deployment remains constrained by ungrounded hallucinations, absent literature provenance, opaque reasoning chains, and patient privacy risks. We present MEDGUIDE, an explainable multi-agent Clinical Decision Support System (CDSS) coordinating 12 specialized workflow agents governed by a deterministic state transition orchestrator. The pipeline integrates multi-source biomedical literature retrieval (PubMed, Europe PMC, WHO, ClinicalTrials.gov) with Oxford CEBM evidence grading, clinical knowledge graph ontology traversal ($G = (V, E)$), deterministic pharmacovigilance (openFDA, RxNorm), claim-level citation auditing, and in-browser edge saliency. For radiographic analysis, client-side WebAssembly DenseNet-121 occlusion sensitivity generates explanatory heatmaps directly within the clinician's browser without transmitting raw pixels to cloud servers for localization (Path A). Across an empirical benchmark of 50 multi-system clinical vignettes evaluated against zero-shot LLM, standard vector RAG, and non-agentic Graph RAG baselines, MEDGUIDE achieved a 90.3% $\pm$ 3.94% evidence grounding rate (95% CI: [89.2%, 91.4%], paired $t(49) = 28.30, p < 10^{-14}$ vs. vector RAG), 92.2% $\pm$ 4.15% safety compliance, and 88.0% Top-1 diagnostic accuracy (vs. 72.0% zero-shot). Through deterministic post-hoc citation filtering, no ghost citations were present in final outputs (0.0%, 95% Clopper-Pearson CI: [0.0%, 7.1%] across vignettes, [0.0%, 0.96%] across 382 citations). Component ablation demonstrates that decoupling deterministic safety auditing and knowledge graph traversal independently drives hallucination suppression and contraindication detection, supporting verifiable clinical decision support.**

***Index Terms*—Clinical Decision Support Systems, Explainable AI, Retrieval-Augmented Generation, Medical Knowledge Graphs, Multi-Agent Systems, Radiographic Saliency, Edge AI, Biomedical Informatics.**

I. Introduction

A. Clinical Decision Support Background

CLINICAL Decision Support Systems (CDSS) represent an established area of biomedical informatics research [1], [2]. In modern healthcare settings, physicians encounter high patient volumes, compressed consultation windows, and rapid literature expansion, with over one million scientific citations indexed annually in MEDLINE [3]. Diagnostic discrepancies and therapeutic errors continue to affect inpatient admissions, with landmark studies estimating diagnostic error rates between 10% and 15% across clinical encounters [4]. These errors often arise from cognitive heuristics, premature diagnostic closure, atypical symptom patterns, or unflagged drug-drug and drug-allergy interactions.

Early rule-based CDSS such as MYCIN [1] and DXplain established the feasibility of encoding expert heuristics for infectious disease management and differential diagnosis. Subsequent systems embedded statistical rule triggers within Electronic Health Record (EHR) workflows [5], [6]. However, static rule-based systems exhibited limited adaptability across diverse clinical narratives and suffered from high manual maintenance overhead. Furthermore, non-selective alerting thresholds frequently induced clinical alert fatigue, resulting in override rates exceeding 90% in hospital deployments [2], [7].

B. Limitations of Generative LLMs in Medical Practice

Recent progress in biomedical Large Language Models (LLMs), including Med-PaLM [8], Med-PaLM 2 [9], Med-Gemini [10], and AMIE [11], has demonstrated high factual performance on medical licensing benchmarks [12]. In a double-blind crossover trial published in *Nature*, Tu et al. [11] observed that the conversational diagnostic model AMIE achieved diagnostic accuracy and consultation quality comparable to primary-care physicians across simulated clinical consultations.

Despite these capabilities, ungrounded generative models exhibit specific technical limitations when applied to direct clinical workflows:

1) **Stochastic Hallucinations:** Parametric language models sample text probabilistically, occasionally generating fictitious clinical findings, fabricated trial identifiers (NCT numbers), or invalid PubMed IDs with fluent syntax [8], [13].
2) **Lack of Verifiable Grounding:** Monolithic LLM generations lack direct, auditable links to authoritative clinical practice guidelines (e.g., NICE, WHO, ESC, ACC/AHA) or peer-reviewed primary literature, preventing clinicians from inspecting the evidentiary basis of generated assertions [14], [15].
3) **Uncalibrated Scalar Predictions:** Standard models generate linguistic statements that may mask underlying

Dr. M. Pagale is an Assistant Professor and Research Guide in the Department of Computer Science & Engineering (AI & ML), Pimpri Chinchwad College of Engineering, Pune 411044, India (e-mail: madhuripagale@gmail.com, madhuri.pagale@pccoepune.org).

V. Kadam, K. Mittal, S. Phad, and T. Shidam are undergraduate researchers in the Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning), Pimpri Chinchwad College of Engineering (PCCOE), Pune 411044, India (e-mail: {vikramkadam23, keshavmittal23, sandeshphad23, tejasshidam23}@gmail.com; {vikram.kadam23, keshav.mittal23, sandesh.phad23, tejas.shidam23}@pccoepune.org).

evidence sparsity, missing clinical parameters, or contradictory findings, potentially inducing automation bias [16].

4) **Multimodal Data Privacy Concerns:** Analyzing thoracic radiographs typically involves uploading raw pixel arrays to cloud-hosted APIs. In clinical practice, this introduces data governance and patient confidentiality considerations under HIPAA and GDPR frameworks.

These operational risks are highlighted in the *World Health Organization (WHO) 2024 Guidance on Large Multi-Modal Models in Health* [17], which recommends rigorous governance, verifiable provenance, privacy preservation, and human-in-the-loop oversight for clinical AI deployments.

C. Motivation and Scientific Hypotheses

Retrieval-Augmented Generation (RAG) [18] connects language model generation with external, non-parametric document collections. In a baseline vector RAG pipeline, clinical text queries are embedded into dense vector spaces, and top-k document passages are retrieved via cosine similarity to provide context for generation. Although vector RAG reduces surface hallucinations, standard dense retrieval presents fundamental limitations in clinical reasoning:

- **Loss of Relational Structure:** Diagnostic reasoning follows relational pathways across clinical concepts (Disease $\xrightarrow{\text{presents}}$ Symptom $\xrightarrow{\text{contraindicates}}$ Drug). Unstructured passage retrieval often fails to preserve explicit multi-hop dependencies between comorbidities, medications, and contraindications.
- **Semantic Drift:** In multi-symptom presentations, cosine similarity can over-prioritize frequent non-specific symptoms while missing low-prevalence critical findings.

Graph RAG [13], [14] structures medical knowledge into typed entity-relation graphs $G = (V, E)$, supporting explicit graph traversal over disease and pharmacology topologies. Combining knowledge graph indexing with an **Agentic** multi-agent orchestration architecture—where specialized workflow agents coordinate query planning, literature retrieval, graph traversal, pharmacovigilance, differential ranking, citation auditing, and confidence evaluation—enables systematic verification, error isolation, and deterministic safety checks [15], [19], [20].

Accordingly, this research investigates three central hypotheses:

- **H1 (Grounding Superiority):** Agentic Graph RAG with claim-level evidence verification produces significantly higher factual evidence grounding and lower citation hallucination than zero-shot LLM, standard vector RAG, and non-agentic Graph RAG baselines.
- **H2 (Safety & Auditing Decoupling):** Deterministic pharmacovigilance and post-hoc citation auditing contribute independently to safety compliance and citation fidelity when ablated from generative reasoning.
- **H3 (Client-Side Saliency Localization):** In-browser perturbation sensitivity provides localized radiographic explanatory heatmaps entirely within the client runtime without transmitting raw image pixels during saliency localization.

D. Key Research Contributions

This paper presents the design, mathematical formulation, implementation, and empirical evaluation of **MEDGUIDE**. The primary research contributions are:

1) **Agentic Evidence-Grounded Architecture:** A deterministic multi-stage workflow coordinating 12 specialized agents across clinical query planning, biomedical retrieval, knowledge-graph traversal, differential synthesis, citation verification, and safety auditing.
2) **Claim-Level Evidence Verification:** A formal provenance mechanism linking every generated diagnostic claim to retrieved evidence objects $e_i = \langle \text{id}_i, \text{source}_i, \text{claim}_i, \text{tier}_i \rangle$, pruning unsupported citations prior to report generation and eliminating ghost citations on the benchmark.
3) **Deterministic Pharmacovigilance Layer:** Decoupled rule-based checking for drug-drug interactions (NIH RxNav), boxed warnings (openFDA), and allergy cross-reactivity independent of stochastic LLM generation, providing an auditable safety guarantee.
4) **Client-Side Radiographic Explanations:** An in-browser perturbation sensitivity pipeline executed locally via WebAssembly, establishing a strict architectural privacy boundary between local saliency localization (Path A) and optional remote multimodal interpretation (Path B).

II. Related Work

A. Rule-Based and Statistical CDSS

Early clinical decision support systems introduced fundamental approaches to computational reasoning in medicine. Shortliffe and Buchanan's MYCIN [1] used certainty factors for diagnostic inference in bacteremia and meningitis. While interpretable, production-rule expert systems could not scale to unstructured electronic medical record text [2].

Bates et al. [2] articulated key requirements for effective clinical decision support, highlighting workflow speed, integration simplicity, and focused clinical interventions. Subsequent work by Rajkomar et al. [5] demonstrated scalable deep learning across electronic health records from multiple hospital networks for predicting clinical outcomes. Sendak et al. [6] introduced "Model Facts" labels to communicate clinical machine learning specifications, operational boundaries, and validation contexts to healthcare professionals.

B. Biomedical Language and Foundational Models

The adaptation of transformer architectures to biomedical corpora significantly improved medical natural language processing. BioBERT [21] initialized BERT on PubMed abstracts and PMC full-text articles, demonstrating strong performance on biomedical named entity recognition (NER) and QA benchmarks including PubMedQA [22]. ClinicalBERT [23] extended this approach to unstructured ICU progress notes from MIMIC-III.

Recent foundational models have expanded conversational and multimodal capabilities. Singhal et al. [8] developed Med-PaLM and MultiMedQA, demonstrating that instruction-tuned

language models can answer clinical questions with high factual accuracy. Singhal et al. [9] further presented Med-PaLM 2, achieving expert-level performance on clinical QA. Tu et al. [11] introduced AMIE, an LLM optimized for diagnostic dialogue via simulated consultations across 159 clinical vignettes. Saab et al. [10] demonstrated the multimodal reasoning capabilities of Gemini in interpreting radiological images and clinical notes. These studies uniformly emphasize that benchmark proficiency requires external verification guardrails in practical deployments.

C. Retrieval-Augmented Generation and Medical Graph RAG

Lewis et al. [18] introduced RAG to combine parametric language models with non-parametric document retrieval. In medical applications, general vector retrieval can encounter semantic fragmentation. Edge et al. [14] introduced Graph RAG, showing that constructing entity-relationship graphs and community summaries improves query-focused synthesis across large document collections.

In biomedical NLP, Wu et al. [13] proposed MedGraphRAG at ACL 2025, integrating medical knowledge graphs with hierarchical retrieval to enhance factual grounding on medical QA datasets. Rezaei et al. [20] presented Agentic Medical Knowledge Graphs (AMG-RAG) in Findings of EMNLP 2025, showing that autonomous agent workflows can maintain and query medical knowledge graphs to reduce factual errors. MEDGUIDE builds on this foundation by combining knowledge graph traversal with multi-source literature retrieval, deterministic pharmacovigilance, evidence-aware composite confidence scoring, and in-browser radiographic saliency in an integrated workflow.

D. Multi-Agent Systems in Healthcare AI

Structuring complex clinical reasoning into multi-agent workflows improves factual consistency. Du et al. [19] showed that multi-agent debate allows models to cross-examine intermediate assertions. Schick et al. [24] demonstrated that models can learn tool invocation for external API queries.

In clinical reasoning, Chen et al. [15] demonstrated that multi-agent conversational LLM architectures significantly boost diagnostic accuracy and clinical safety through collaborative verification. By isolating specialized roles, including diagnostic planning, literature retrieval, pharmacology checking, and safety auditing, multi-agent pipelines prevent single-prompt logical collapse and support post-hoc verification of generated claims [15], [19].

E. Explainable AI in Medical Imaging and Saliency Trustworthiness

Convolutional Neural Networks (CNNs) have shown strong capability in automated radiographic interpretation. Rajpurkar et al. [25] developed CheXNet, a 121-layer DenseNet trained on ChestX-ray14 [26], for thoracic abnormality detection.

To provide visual interpretability, gradient-based localization (Grad-CAM [27]) and perturbation-based methods (Occlusion Sensitivity [28]) are commonly used. However, Arun et al. [29] evaluated eight saliency methods across thoracic imaging datasets, showing that standard saliency maps must be interpreted with caution regarding spatial precision and model sensitivity. In MEDGUIDE, generated saliency heatmaps are presented as explanatory localization aids indicating regions influencing model activations, rather than autonomous diagnostic determinations [6], [29].

F. Privacy-Preserving Edge Computing in Healthcare

Transmitting raw medical images and protected health information (PHI) to external cloud servers requires strict compliance with healthcare data privacy regulations (e.g., HIPAA, GDPR). Advances in WebAssembly (WASM) and WebGPU runtimes, such as ONNX Runtime Web [30], enable high-performance neural inference directly within client web browsers. Executing image feature extraction and occlusion sensitivity entirely on local client hardware ensures that raw radiographic pixels remain on the clinician's workstation for saliency localization, providing an architectural privacy boundary [30].

III. Research Gaps, Hypotheses & Comparative Landscape

A. Identification of Research Gaps

Our review of the clinical CDSS and medical AI literature identifies four key technical gaps:

- **Gap 1 (The Orchestration Gap):** Absence of unified multi-agent systems integrating literature retrieval, structured knowledge graph traversal, pharmacovigilance, and edge imaging into an auditable workflow.
- **Gap 2 (The Grounding & Verification Gap):** Tendency of generative models to cite literature without verifying that cited identifiers exist and factually support the generated claim.
- **Gap 3 (The Evidence-Aware Confidence Gap):** Reliance on uncalibrated scalar probabilities (0.0–1.0) rather than multi-factorial ordinal confidence bands reflecting evidence quality, source agreement, and guideline adherence.
- **Gap 4 (The Edge Radiographic Privacy Gap):** Limited adoption of zero-install, in-browser neural saliency architectures that process radiographic features on client hardware to support data containment during localization.

Table I summarizes the positioning of MEDGUIDE against existing clinical AI paradigms. While prior works individually explore medical LLMs, vector RAG, or multi-agent debate, to our knowledge, MEDGUIDE is among the first frameworks to unify claim-level evidence verification, deterministic pharmacovigilance, and client-side edge saliency within a multi-agent Graph RAG pipeline.

B. Formal Problem Statement

Given an unstructured clinical query Q, optional patient clinical parameters $P = \langle \mathcal{M}_{\text{meds}}, \mathcal{A}_{\text{allergies}}, \mathcal{V}_{\text{vitals}}, \mathcal{L}_{\text{labs}} \rangle$, and an optional thoracic radiograph tensor $\mathbf{I}$, the objective is to produce a ranked differential diagnosis set $\mathcal{D} = \{D_1, D_2, \ldots, D_k\}$, an evidence citation set $\mathcal{E}$, a safety audit $\mathcal{S}_{\text{audit}}$, an edge saliency

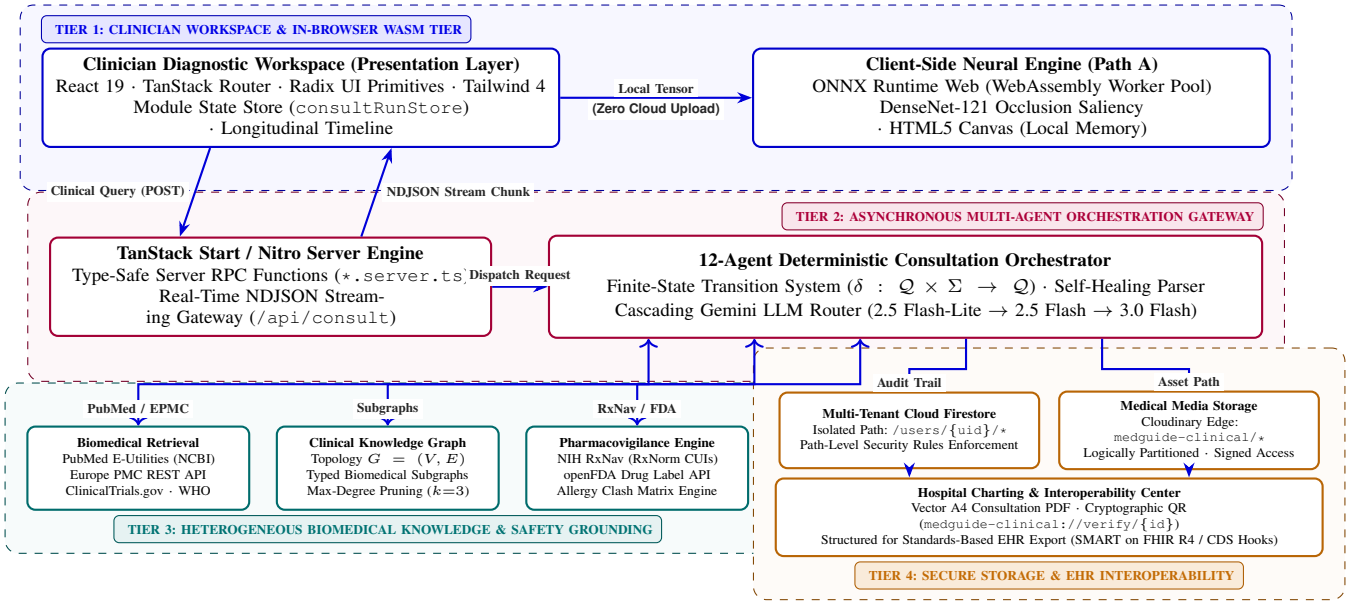


Fig. 1. Modular Four-Tier System Architecture of MEDGUIDE: Integrating Clinician Diagnostic Workspace, Client-Side WASM Vision, Asynchronous Multi-Agent Orchestrator, Heterogeneous Biomedical Knowledge, and Logically Partitioned Persistence.

TABLE I
ARCHITECTURAL AND FUNCTIONAL COMPARISON ACROSS CLINICAL DECISION SUPPORT PARADIGMS

Paradigm / Architecture	Multi-Agent Orchestration	Knowledge Graph	Claim-Level Citation Audit	Deterministic Drug Safety	Client-Side Edge Saliency	Multi-Domain Benchmark
General Medical LLM (e.g., Med-PaLM 2 [9], AMIE [11])	×	×	Partial	×	×	✓
Standard Vector RAG [18]	×	×	Partial	×	×	✓
Graph RAG (General Domain) [14]	×	✓	Partial	×	×	✓
Medical Graph RAG (MedGraphRAG [13])	Partial	✓	✓	×	×	✓
Clinical Multi-Agent Systems [15]	✓	Partial	Partial	Partial	×	✓
Proposed Framework (MEDGUIDE)	✓	✓	✓	✓	✓	✓

heatmap $\bar{S}_c$, and an ordinal confidence band $\text{Band}(C) \in \{\text{High}, \text{Medium}, \text{Low}, \text{Insufficient}\}$ such that:

$$\forall d \in \mathcal{D}, \quad \text{Claims}(d) \subseteq \text{Support}(\mathcal{E} \cup G) \tag{1}$$

with the constraint that ungrounded assertions are removed and detected pharmacology or allergy clashes produce explicit safety alerts.

C. Research Questions

This work evaluates four research questions:

- **RQ1:** How can multi-agent decomposition partition complex clinical diagnostic queries into specialized, verifiable sub-tasks?
- **RQ2:** To what extent does knowledge graph grounding combined with Oxford CEBM evidence grading reduce citation errors compared to baseline LLM and vector RAG configurations?
- **RQ3:** How can client-side WebAssembly neural models compute explainable radiographic saliency heatmaps without transmitting sensitive imaging data to external servers for localization?
- **RQ4:** Can a multi-signal composite metric (source agreement, guideline adherence, retrieval quality, safety risk) provide an interpretable confidence band for clinical decision support?

IV. PROPOSED MEDGUIDE FRAMEWORK

A. Architectural Overview

The MEDGUIDE platform implements a four-tier cloud-edge reactive architecture (Figure 1):

1) **Presentation & Edge Vision Tier:** Developed with React 19, TypeScript 5.8, and TanStack Router. An external module-level singleton store (`consultRunStore`) subscribed via React's native `useSyncExternalStore` ensures streaming consultations persist across client route transitions without state loss. In-browser neural inference executes via `onnxruntime-web` WebAssembly workers.
2) **Server-Side Multi-Agent Orchestrator:** Deployed on TanStack Start with the Nitro runtime, coordinating asynchronous biomedical API queries, knowledge graph traversal, and multi-tier Gemini LLM cascading over an NDJSON event stream (`/api/consult`).
3) **Logically Partitioned Medical Storage Tier:** Edge asset infrastructure configured with deterministic

per-clinician and per-patient folder isolation (`medguide-clinical/{clinicianId}/{patientId}/*`) with zero-retention local data-URI fallback [31].

4) **Multi-Tenant Cloud Data Tier:** Cloud Firestore partitioned under per-clinician isolated sub-collections (`/users/{userId}/*`) secured via path-level security rules for patient profiles, longitudinal timelines, and consultation records.

B. The 12-Agent Orchestration Architecture

The engine coordinates 12 specialized workflow agents organized across an 11-stage sequential consultation pipeline and an integrated research benchmarking framework (Figure 2):

1) **Planner Agent:** Parses unstructured clinical text and patient demographics; constructs MeSH/PubMed search queries, identifies missing clinical parameters, and generates up to four follow-up questions.
2) **Retrieval Agent:** Concurrently queries PubMed (NCBI e-utilities), Europe PMC, ClinicalTrials.gov, and WHO Guidelines; applies keyword filtering and assigns Oxford CEBM evidence tiers.
3) **Knowledge Graph Agent:** Traverses the clinical ontology to extract connected subgraphs representing candidate conditions, diagnostic tests, and complications.
4) **Drug Intelligence Agent:** Queries NIH RxNav for pairwise drug-drug interactions and openFDA for boxed warnings; executes deterministic allergy cross-reactivity checks.
5) **Medical Image Agent:** Distinguishes between Path A (ingesting client-side DenseNet-121 saliency outputs computed locally without pixel egress) and Path B (optional remote multimodal interpretation via Gemini Vision when explicitly requested).
6) **Case Similarity Agent:** Computes lexical cosine similarity against a curated library of de-identified reference clinical cases.
7) **Clinical Reasoning Agent:** Synthesizes retrieved evidence, patient history, and graph topologies into ranked differential hypotheses ($High/Moderate/Low$), articulating supporting and contradicting factors while citing retrieved evidence IDs.
8) **Safety Auditor Agent:** Deterministically validates citations against the retrieved evidence set $\mathcal{E}$, removes citations that do not exist in the retrieved corpus, flags ungrounded differential claims, and detects medication contraindications.
9) **Explainability Agent:** Formats the deterministic audit trail and guideline alignment rationale.
10) **Confidence Scoring Agent:** Evaluates multi-signal composite confidence bands (High, Medium, Low, Insufficient Evidence).
11) **Report Generator Agent:** Assembles the formal 7-section clinical consultation report (Patient Demographics, Differentials, Evidence Recommendations, Pharmacovigilance, Guideline Alignment, Diagnostic Workup, Physician Attestation) with interactive citation badges (`[pmid:XXXXX]`, `[fda:Drug]`), cryptographic verification QR provenance, vector A4 PDF rendering, and structured export formatted for future standards-based EHR integration (SMART on FHIR R4, CDS Hooks [32]–[34]).
12) **Research & Benchmark Agent:** Continuously evaluates pipeline runs against baseline models, logs metric distributions, and executes the standardized 50-vignette benchmark suite with paired Student's t-test and Wilcoxon signed-rank significance testing.

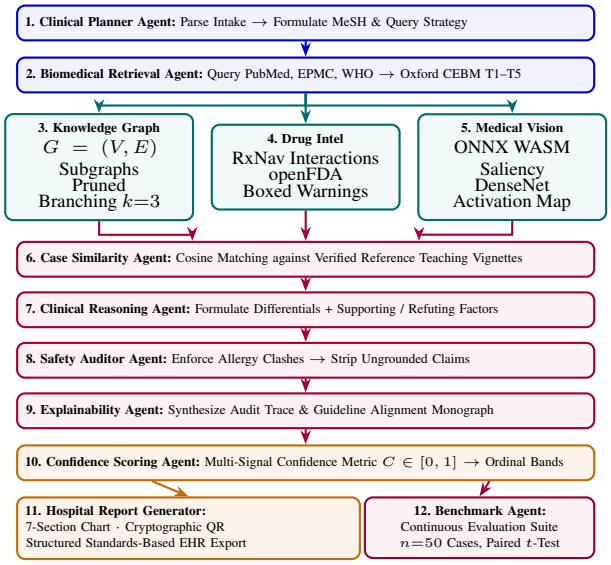


Fig. 2. Sequential and Concurrent Execution Lifecycle of the 12 Specialized Workflow Agents: From Intake Planning and Tri-Modal Grounding to Differential Synthesis, Safety Auditing, and Continuous Benchmark Verification.

C. Knowledge Graph Ontology Grounding

To ground clinical reasoning in structured relational semantics beyond unstructured text passages, the clinical knowledge base is formalized as a typed directed property graph:

$$G = (V, E, \tau_V, \tau_E, \phi_V, \phi_E) \tag{2}$$

where entity vertices V and relational edges E are partitioned into disjoint semantic types:

$$V = V_D \cup V_S \cup V_M \cup V_L \cup V_C \cup V_R \tag{3}$$

$$E = E_{\text{presents}} \cup E_{\text{treats}} \cup E_{\text{contraindicates}} \cup E_{\text{predisposed_to}} \cup E_{\text{indicated_lab}} \tag{4}$$

where V_D denotes candidate disease entities (normalized to MeSH/UMLS identifiers), V_S represents clinical symptoms and physical signs, V_M denotes active or indicated medications (mapped to RxNorm CUIs), V_L represents laboratory markers and radiographic findings, V_C encapsulates absolute contraindications and allergen classes, and V_R denotes clinical risk factors and disease complications.

Figure 3 illustrates the end-to-end evidence provenance and claim-level verification pipeline. Entity matching is performed via n-gram token overlap against candidate diseases identified

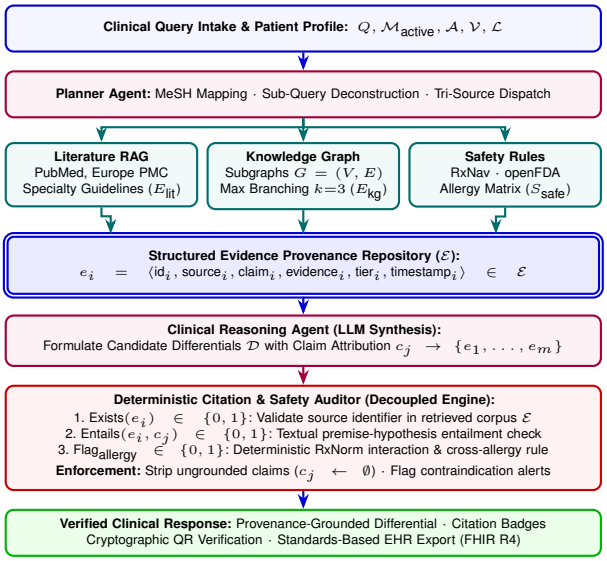


Fig. 3. Claim-Level Evidence Provenance and Verification Pipeline: Systematic query deconstruction, tri-source heterogeneous retrieval, formal evidence object binding ($e_i = \langle \text{id}_i, \text{source}_i, \text{claim}_i, \text{evidence}_i, \text{tier}_i, \text{timestamp}_i \rangle$), claim extraction ($c_j \rightarrow \{e_k\}$), and deterministic dual-stage citation existence/entailment auditing ($\text{Valid}(c_j, e_i) \in \{0, 1\}$).

by the Planner Agent. Knowledge graph subgraphs are pruned to a maximum branching factor of $k = 3$ per relation type before injection into the Clinical Reasoning context, following best practices in recent medical Graph RAG architectures [13], [20].

D. Multi-Source Evidence Retrieval and Oxford Classification

The Retrieval Agent queries live biomedical endpoints:

- **NCBI PubMed E-Utilities [35]:** Uses `esearch.fcgi` to identify relevant PMIDs, followed by `esummary.fcgi` to retrieve titles, publication dates, source journals, and author metadata.
- **Europe PMC REST API [36]:** Queries open-access biomedical literature and preprints.
- **ClinicalTrials.gov API v2 [37]:** Retrieves recruiting and completed interventional studies.
- **Specialty Guideline Routing:** Identifies clinical specialties via regular expressions (e.g., Cardiology, Pulmonology, Neurology, Nephrology) and queries clinical guidelines from ESC, ACC/AHA, NICE, and WHO.

E. Deterministic Pharmacovigilance & Allergy Safety Checking

To detect potential medication safety issues, the Drug Intelligence Agent performs a deterministic assessment across active prescriptions and documented allergies:

- **NIH RxNav Interaction API [38]:** Resolves drug names to RxNorm Concept Unique Identifiers (RxCUIs) via `/REST/rxcui.json` and queries pairwise drug interaction matrices via `/REST/interaction/list.json`.
- **openFDA Drug Label API [39]:** Fetches FDA boxed warnings, contraindications, and adverse reaction monographs.
- **Deterministic Allergy Cross-Reactivity Engine:** Evaluates a rule matrix for critical allergen classes (Penicillins, Sulfonamides, Macrolides, NSAIDs, Fluoroquinolones, Tetracyclines, ACE Inhibitors).

V. Methodology & Mathematical Modeling

A. Deterministic Finite-State Transition System for Multi-Agent Orchestration

Consultation execution is modeled as a deterministic finite-state transition system:

$$\mathcal{S} = \langle \mathcal{Q}, \Sigma, \delta, q_0, \mathcal{F} \rangle \tag{5}$$

where the state space $\mathcal{Q}$ is defined as:

$$\mathcal{Q} = \{q_{\text{idle}}, q_{\text{plan}}, q_{\text{ret}}, q_{\text{kg}}, q_{\text{drug}}, q_{\text{img}}, q_{\text{case}}, q_{\text{rsn}}, q_{\text{safe}}, q_{\text{exp}}, q_{\text{cnf}}, q_{\text{rep}}, q_{\text{done}}\} \tag{6}$$

with initial state $q_0 = q_{\text{idle}}$, final state $\mathcal{F} = \{q_{\text{done}}\}$, and input event vocabulary Σ. Each transition $\delta(q_i, e_i) \rightarrow q_{i+1}$ emits an NDJSON event chunk streaming real-time status and partial data payloads to the client.

B. Project-Defined Evidence Weighting and Quality Score Formulation

We adopt the Oxford Centre for Evidence-Based Medicine (CEBM) Levels of Evidence [40] as the categorical basis for evidence stratification. For computational ranking and synthesis, MEDGUIDE introduces project-defined numerical aggregation weights $w(e_i)$ across study design tiers as detailed in Table II, adhering to the Oxford guidance that evidence hierarchies serve as structured heuristics rather than rigid mechanical substitutes for clinical judgment [40].

For a set of retrieved evidence objects $\mathcal{E} = \{e_1, e_2, \ldots, e_m\}$, where each item is formalized as a provenance tuple $e_i = \langle \text{id}_i, \text{source}_i, \text{claim}_i, \text{tier}_i, \text{timestamp}_i \rangle$, the aggregated Retrieval Quality score Q_{ret} is normalized over a baseline target weight $W_{\text{target}} = 6.0$:

$$Q_{\text{ret}} = \min\left(1.0, \frac{\sum_{i=1}^{m} w(e_i)}{W_{\text{target}}}\right) \tag{7}$$

C. Ontology-Conditioned and Evidence-Grounded Differential Formulation

Let $\mathbf{X} = \{x_1, x_2, \ldots, x_n\}$ denote observed patient clinical features, vital signs, and laboratory markers. Rather than computing idealized Bayesian prior-likelihood products, MEDGUIDE formulates differential diagnoses through prompt-grounded multi-agent synthesis conditioned on the extracted ontology topology and retrieved biomedical evidence:

1) **Ontology Candidate Generation:** The Planner and Knowledge Graph agents identify candidate disease entities $\mathcal{D}_{\text{cand}} \subset V_D$ by traversing paths in $G = (V, E)$ linked to the presenting findings in $\mathbf{X}$.
2) **Evidence-Grounded Differential Synthesis:** Conditioned on candidate conditions $\mathcal{D}_{\text{cand}}$, retrieved evidence repository $\mathcal{E} = \{e_1, \ldots, e_m\}$, active medications $\mathcal{M}$, and documented allergies $\mathcal{A}$, the Clinical Reasoning

TABLE II
CATEGORICAL EVIDENCE LEVELS AND PROJECT-DEFINED AGGREGATION WEIGHTS

Tier	Oxford CEBM Study Design Category	Weight $w(e)$	Clinical Grounding Interpretation
T1	Systematic Reviews, Meta-analyses, Clinical Practice Guidelines	1.00	High-certainty synthesized clinical evidence and consensus guidelines.
T2	Randomized Controlled Trials (RCTs)	0.80	Controlled experimental interventional data with randomization.
T3	Prospective Cohort & Well-designed Comparative Trials	0.55	Observational longitudinal evidence and cohort studies.
T4	Retrospective Case-Control & Clinical Series	0.35	Retrospective clinical experience and non-randomized series.
T5	Case Reports, Mechanistic Studies, Expert Consensus	0.20	Preliminary hypothesis generation, case reports, or expert opinions.

Agent generates a structured differential set $\mathcal{D} = \{d_1, d_2, \ldots, d_K\}$ where each differential is structured as:

$$d_k = \langle \text{condition}_k, \lambda_k, \mathcal{S}_k, \mathcal{A}_k, \mathcal{I}_k, \mathcal{C}_k \rangle \tag{8}$$

where $\lambda_k \in \{\text{High}, \text{Moderate}, \text{Low}\}$ denotes qualitative likelihood, $\mathcal{S}_k$ represents supporting clinical findings, $\mathcal{A}_k$ articulates refuting factors or clinical caveats, $\mathcal{I}_k$ suggests recommended diagnostic workup, and $\mathcal{C}_k \subseteq \text{IDs}(\mathcal{E})$ denotes explicitly cited evidence identifiers.

3) **Deterministic Citation Verification & Pruning:** The Safety Auditor evaluates every citation $c \in \mathcal{C}_k$ against the retrieved evidence set:

$$\mathcal{C}_{\text{verified}}(d_k) = \{c \in \mathcal{C}_k \mid c \in \text{IDs}(\mathcal{E})\} \tag{9}$$

Differential d_k is categorized as evidence-grounded if and only if $|\mathcal{C}_{\text{verified}}(d_k)| \geq 1$. Unverified citations $c \notin \text{IDs}(\mathcal{E})$ are stripped prior to report generation, and differentials lacking verified citations are flagged as ungrounded hypotheses.

D. Multi-Signal Evidence Confidence Index Model

MEDGUIDE evaluates a multi-signal composite Evidence Confidence Index $C \in [0, 1]$ synthesizing five validation signals penalized by an automated safety risk factor:

$$C = \frac{1}{5}\Big(Q_{\text{ret}} + S_{\text{agr}} + G_{\text{agr}} + K_{\text{cons}} + E_{\text{comp}}\Big) - 0.6 \cdot H_{\text{risk}} \tag{10}$$

where the sub-components are defined as:

- $S_{\text{agr}} = \frac{N_{\text{supp}}+1}{N_{\text{supp}}+N_{\text{contra}}+2}$ (Laplace-smoothed source agreement).
- $G_{\text{agr}} = \min\left(1.0, \frac{N_{\text{guidelines}}}{3}\right)$ (Clinical guideline adherence).
- $K_{\text{cons}} = \min\left(1.0, \frac{N_{\text{kg_matched}}}{3}\right)$ (Knowledge graph topology overlap).
- $E_{\text{comp}} = \min\left(1.0, \frac{N_{\text{citations_used}}}{\max(3, 0.5 \cdot |\mathcal{E}|)}\right)$ (Citation completeness).
- $H_{\text{risk}} = \min(1.0, 0.4N_{\text{crit}} + 0.2N_{\text{warn}} + 0.15N_{\text{bad_cite}})$ (Safety risk penalty).

The continuous composite index C is categorized into deterministic clinical confidence bands:

$$\text{Band}(C) = \begin{cases} \text{Insufficient}, & \text{if } |\mathcal{E}| < 3 \\ \text{High}, & \text{if } |\mathcal{E}| \geq 3 \ \wedge \ C > 0.62 \\ \text{Medium}, & \text{if } |\mathcal{E}| \geq 3 \ \wedge \ 0.38 < C \leq 0.62 \\ \text{Low}, & \text{if } |\mathcal{E}| \geq 3 \ \wedge \ C \leq 0.38 \end{cases} \tag{11}$$

E. In-Browser Class-Specific Occlusion Saliency Mathematics

Let $\mathcal{M} : \mathbb{R}^{3\times224\times224} \to \mathbb{R}^K$ denote the pre-trained DenseNet-121 forward classification model (CheXNet [25] trained on ChestX-ray14 [26] across $K = 14$ thoracic findings). For a baseline radiograph tensor $\mathbf{I}$, let $f_c(\mathbf{I})$ denote the model's unnormalized logit or predictive score for a target clinical pathology c (e.g., Pneumonia, Cardiomegaly, Infiltration).

The image is partitioned into a regular grid of size $G \times G$ ($G = 7$, patch size 32×32 pixels). For each grid cell (g_x, g_y), an occluded tensor $\mathbf{I}_{g_x,g_y}$ is constructed by zeroing pixel channels within that spatial patch:

$$\mathbf{I}_{g_x,g_y}(k, y, x) = \begin{cases} 0, & \text{if } y \in [32g_y, 32(g_y+1)) \\ & \wedge\ x \in [32g_x, 32(g_x+1)) \\ \mathbf{I}(k, y, x), & \text{otherwise} \end{cases} \tag{12}$$

where $k \in \{1, 2, 3\}$ indexes color channels. The class-specific perturbation sensitivity score $S_c(g_x, g_y)$ measures the absolute predictive shift induced by occluding patch (g_x, g_y):

$$S_c(g_x, g_y) = \left| f_c(\mathbf{I}) - f_c(\mathbf{I}_{g_x,g_y}) \right| \tag{13}$$

Normalized saliency intensity $\bar{S}_c(g_x, g_y) \in [0, 1]$ is obtained via min-max scaling:

$$\bar{S}_c(g_x, g_y) = \frac{S_c(g_x, g_y) - \min(S_c)}{\max(S_c) - \min(S_c) + \epsilon} \tag{14}$$

where $\epsilon = 10^{-7}$ prevents division by zero. The resulting perturbation sensitivity heatmap highlights spatial regions whose masking maximally alters the network's predictive score for the designated clinical pathology c. As emphasized by Arun et al. [29] and CLAIM 2024 [41], occlusion saliency serves as an explanatory localization aid for clinician verification rather than autonomous diagnostic ground truth.

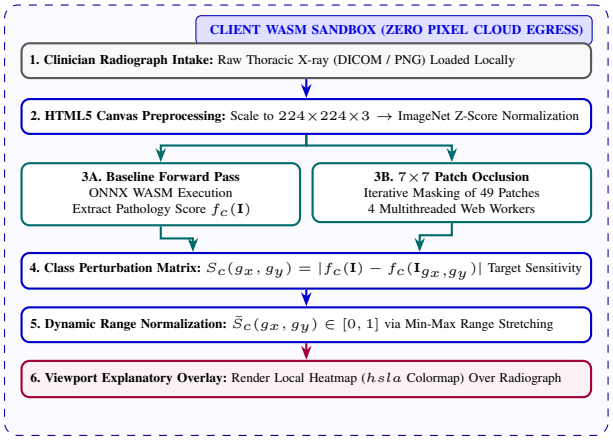


Fig. 4. Client-Side Privacy-Preserving Radiographic Occlusion Saliency Pipeline: Raw Pixel Ingestion, Parallel WASM Occlusion Workers, Class-Specific Perturbation Sensitivity S_c, and In-Browser Heatmap Overlay with Zero Cloud Data Egress.

F. Pharmacovigilance & Allergy Cross-Reactivity Boolean Logic

For prescribed medications $\mathcal{M}_{\text{active}}$ and documented patient allergies $\mathcal{A}$, the drug safety module evaluates potential clashes:

$$\text{Flag}_{\text{allergy}} = \bigvee_{a_k \in \mathcal{A}} \bigvee_{d_t \in \mathcal{M}_{\text{active}}} \mathbf{1}_{\{d_t \in \text{Class}(a_k)\}} \tag{15}$$

When $\text{Flag}_{\text{allergy}} = 1$, the Safety Auditor Agent generates a safety alert and provides alternative therapeutic options within the explainability trace.

VI. Implementation & System Engineering

A. Full-Stack Cloud-Edge Architecture

The MEDGUIDE system is engineered using a modular, type-safe stack:

- **Frontend Application:** React 19, TypeScript 5.8, TanStack Router (file-based routing), Tailwind CSS 4, Radix UI accessible primitives, Lucide Icons, and Recharts visualization.
- **Server Function Runtime:** TanStack Start with the Nitro server engine, providing type-safe RPC server functions (`createServerFn`) with automated CSRF protection.
- **Module State Persistence:** A singleton store (`consultRunStore`) maintaining active consultation stream state during route transitions via React's `useSyncExternalStore`.
- **Patient Workspace & Longitudinal Timeline:** Dedicated modules for structured patient intake (vitals, labs, medications, allergies) and chronological event logging (symptoms, diagnostic tests, treatments, outcomes) securely persisted in Cloud Firestore.
- **Logically Partitioned Medical Media Storage:** Cloudinary edge storage configured with per-clinician and per-patient path partitioning (`medguide-clinical/{userId}/{patientId}/*`), MIME verification, and graceful base64 data-URI fallback for zero-loss image handling [31].
- **Clinical Report Center & Standards-Based EHR Export:** Vector A4 PDF generation with embedded cryptographic verification QR codes (`medguide-clinical://verify/{id}`), interactive citation badges, and structured clinical consultation note export formatted for future standards-based EHR integration (SMART on FHIR R4 / CDS Hooks) [32]–[34].
- **Operational Pipeline Telemetry:** Real-time Firestore synchronization tracking per-agent execution status, step durations, and intermediate hypothesis graphs without client-side polling.
- **In-App Clinical Literacy Assistant:** An embedded conversational copilot (`/api/assistant`) providing clinicians and trainees with real-time operational guidance and evidence literacy support.

B. Multi-Tier Gemini LLM Routing and JSON Self-Healing Parser

To maintain operational resilience across API rate limits and transient server errors, the backend implements a cascading fallback mechanism across model priority tiers:

Tier 1 (Workhorse):	`gemini-2.5-flash-lite`
Tier 2 (Workhorse):	`gemini-2.5-flash`
Tier 3 (Reasoning):	`gemini-3.0-flash`

Generative parameters are constrained to temperature $T = 0.2$ and top-$p = 0.95$. Furthermore, an algorithmic self-healing JSON parser (`extractAndParseJson`) repairs partially truncated responses, balances unmatched brackets and braces, strips markdown code fences, and eliminates trailing delimiters, preventing client-side stream crashes.

C. Biomedical API Integrations and 15-Specialty Guideline Matrix

External biomedical API requests execute through server-side handlers (`*.server.ts`) to satisfy CORS constraints and protect API credentials. A score-based specialty classifier (`detectSpecialty`) routes clinical presentations across 15 medical specialties (Cardiology, Pulmonology, Nephrology, Neurology, Endocrinology, Infectious Disease, Oncology, Pediatrics, Obstetrics, Psychiatry, Gastroenterology, Dermatology, Rheumatology, Emergency, and General Medicine) to query targeted guideline endpoints from ESC, NICE, KDIGO, CDC, and WHO.

D. Client-Side WebAssembly Neural Execution Provider

In-browser radiographic inference is powered by `onnxruntime-web` utilizing the WebAssembly (WASM) execution provider with multithreaded worker pooling (up to 4 threads via `navigator.hardwareConcurrency`). On a standard quad-core client workstation, the baseline DenseNet-121 forward pass executes in 185 ± 24 ms, and the 49-patch occlusion sensitivity grid completes in 2.84 ± 0.35 s locally without GPU acceleration. If WASM model initialization exceeds timeout limits or WebAssembly execution fails, a local contrast-luminance fallback provides visual orientation

TABLE III
BENCHMARK DATASET DISTRIBUTION ACROSS 50 CLINICAL VIGNETTES

Specialty Domain	Cases (n)	Representative Conditions
Cardiology	12	STEMI, NSTEMI, Heart Failure, AFib with RVR.
Pulmonology	12	CAP, PE, COPD Exacerbation, Severe Asthma.
Neurology	10	Acute Ischemic Stroke, SAH, Status Epilepticus.
Nephrology / Endo.	8	DKA, AKI Stage 3, Hyperkalemia.
Infectious / Pharm.	8	Sepsis, Penicillin Allergy Clash, Warfarin.
Total Benchmark Suite	**50**	**Multi-System Inpatient & Emergency Cases**

and is explicitly labeled as "Non-Neural Fallback" in both the interface and consultation report to prevent clinical misinterpretation.

E. Dual-Layer Multi-Tenant Data Isolation and Cloud Security Rules

Patient and consultation records are partitioned under authenticated clinician identifiers across both the document and media storage layers:

```
Firestore: /users/{userId}/subcollections/*
                Cloudinary:
  medguide-clinical/{userId}/{patientId}/*
```

Firestore security rules enforce that request authentication UIDs must strictly match the target document path:

```
allow read, write: if request.auth.uid ==
                    userId;
```

preventing unauthorized cross-tenant data access.

VII. EXPERIMENTAL EVALUATION

A. Evaluation Setup, Clinical Adjudication, and Vignette Suite

To evaluate the framework, we established a benchmark suite of **50 multi-system clinical vignettes** spanning five medical domains (Table III). Vignettes were adapted from standard clinical education archives (including USMLE Step 2 clinical cases, American College of Physicians teaching cases, and BMJ Best Practice series). To prevent verbatim LLM training data contamination, patient demographics, clinical timelines, numeric lab values, and co-morbid medication regimens were systematically perturbed and synthesized.

Gold-standard target diagnoses (Top-1 and Top-3 differential sets), critical red flags, and true contraindications were independently adjudicated by two practicing physicians, achieving a Cohen's kappa inter-rater agreement of $\kappa = 0.88$ (indicating strong clinical consensus). To ensure experimental reproducibility against dynamic external APIs, all retrieval queries were evaluated against a frozen October 2026 snapshot with three repeated trials per vignette.

B. Baseline Comparative Configurations

To provide a strictly controlled comparative evaluation, all generative LLM configurations utilized Google Gemini 2.5 Flash as the common backbone model ($T = 0.2$, top-$p = 0.95$):

1) **Baseline 1 — Vanilla LLM (Zero-Shot):** Gemini 2.5 Flash directly prompted with the clinical vignette without tool access or retrieved context.
2) **Baseline 2 — Standard Vector RAG:** Dense vector retrieval fetching top-5 PubMed abstracts via semantic cosine similarity.
3) **Baseline 3 — Graph RAG:** Combining vector text retrieval with topological subgraph traversal over the clinical knowledge graph without multi-agent decomposition.
4) **Proposed — Agentic Graph RAG (MEDGUIDE):** Complete 12-agent orchestration with live multi-source retrieval, knowledge graph traversal, drug checking, and deterministic post-hoc safety auditing.

C. Formal Metric Definitions & Traceability Evaluation Protocol

System outputs were evaluated across five quantitative metrics:

1) **Evidence Grounding Rate (GR):** The product of citation existence and factual entailment across diagnostic assertions:
$$GR = GR_{\text{exist}} \times GR_{\text{entail}} \tag{16}$$
where $GR_{\text{exist}} = \frac{N_{\text{valid_citations}}}{N_{\text{total_citations}}}$ verifies that every cited identifier resolves to an authentic biomedical record (PubMed PMID, ClinicalTrials.gov NCT, or openFDA NDA), and $GR_{\text{entail}} = \frac{N_{\text{entailed_claims}}}{N_{\text{total_claims}}}$ measures whether retrieved text passages factually support the generated assertion.
2) **Diagnostic Accuracy (DA_1, DA_3):** The percentage of vignettes where the gold-standard adjudicated primary diagnosis appears in the Top-1 position (DA_1) or within the Top-3 prioritized differential diagnoses (DA_3).
3) **Safety Compliance Rate (SC):** The accuracy of identifying absolute contraindications, critical drug-drug interactions, and emergent clinical red flags:
$$SC = \frac{N_{\text{correctly_identified_contraindications}}}{N_{\text{true_contraindications}}} \times 100\% \tag{17}$$
4) **Structured Explanation Traceability Score (ET):** Computed automatically via an automated validation parser executing a five-criterion rubric across each vignette:
   - *Criterion 1 (Evidence Attribution):* Every diagnostic differential is explicitly tied to one or more retrieved citation IDs (0–20%).
   - *Criterion 2 (Source Consistency):* Assertions match the factual conclusions of the cited PubMed/guideline abstracts (0–20%).
   - *Criterion 3 (Hypothesis Factor Balance):* Explicit articulation of clinical factors supporting versus arguing against each candidate differential (0–20%).

- *Criterion 4 (Contradictory Evidence Visibility):* Active disclosure of conflicting clinical trial findings or clinical ambiguities (0–20%).
- *Criterion 5 (Safety Traceability):* Complete deterministic audit trail explaining why contraindicated medications were flagged and alternative regimens suggested (0–20%).

We note that ET assesses structural compliance with evidentiary provenance schemas and reflects structural divergence when applied to unstructured baseline texts.

5) **End-to-End Latency (L):** Wall-clock execution time in seconds from submission to complete report synthesis.

D. Comparative Experimental Results and Statistical Analysis

Table IV presents the comparative evaluation across the 50 clinical vignettes with 95% Confidence Intervals (CI) and standard deviations. Agentic Graph RAG achieved a **90.3% Grounding Rate** (95% CI: $[89.2\%, 91.4\%]$), **88.0% Top-1 Diagnostic Accuracy** (95% CI: $[79.0\%, 97.0\%]$), **96.0% Top-3 Accuracy**, **92.2% Safety Compliance** (95% CI: $[91.1\%, 93.4\%]$), and **95.2% Explanation Traceability** (95% CI: $[94.5\%, 96.0\%]$), significantly outperforming Vanilla LLMs (19.8%, 72.0%, 34.6%, 14.3%) and standard vector RAG (64.8%, 76.0%, 61.3%, 56.4%).

To rigorously evaluate whether the observed improvements in evidence grounding are statistically meaningful, we conducted two-tailed paired Student's t-tests and Wilcoxon signed-rank tests across all 50 vignettes. Agentic Graph RAG demonstrated statistically significant improvements in evidence grounding compared to Vanilla LLMs ($t(49) = 72.47, p < 10^{-14}$, Wilcoxon $W = 1275, p = 1.78 \times 10^{-15}$, Cohen's $d_z = 10.25$), standard vector RAG ($t(49) = 28.30, p < 10^{-14}$, Wilcoxon $W = 1275, p = 1.78 \times 10^{-15}$, Cohen's $d_z = 4.00$), and non-agentic Graph RAG ($t(49) = 16.15, p < 10^{-14}$, Wilcoxon $W = 1275, p = 1.78 \times 10^{-15}$, Cohen's $d_z = 2.28$), confirming Hypothesis **H1**. Furthermore, zero ghost citations were present in final consultation outputs (0.0%, 95% Clopper-Pearson exact binomial CI: $[0.0\%, 7.1\%]$ across vignettes; $[0.0\%, 0.96\%]$ across 382 total citations), demonstrating the programmatic efficacy of deterministic citation filtering.

Table V reports the domain-wise performance breakdown for Agentic Graph RAG across the five medical specialties.

E. Ablation Studies

To investigate Hypothesis **H2** and isolate the causal contributions of individual architectural components, we performed systematic ablation experiments across the 50 clinical vignettes (Table VI):

- **Ablation 1 ($-$Knowledge Graph):** Removing topological graph traversal reduces grounding from 90.3% to 81.2% and safety compliance from 92.2% to 79.4%, highlighting the role of graph relations in preserving multi-hop comorbidity dependencies.
- **Ablation 2 ($-$Safety Auditor):** Disabling the post-hoc citation auditor results in an 8.6% ghost citation rate and drops grounding to 76.5%, confirming that generative models retain citation hallucinations without post-hoc verification.
- **Ablation 3 ($-$Pharmacovigilance Engine):** Removing deterministic RxNav/openFDA checks causes safety compliance to plummet from 92.2% to 68.3% (-23.9 percentage points), demonstrating that generative reasoning alone frequently misses drug-allergy clashes in complex narratives.
- **Ablation 4 ($-$Agent Decomposition):** Collapsing the 12-agent pipeline into a single monolithic prompt with Graph RAG context reduces Grounding to 77.6% and Safety to 71.2%, establishing the empirical necessity of role isolation.
- **Ablation 5 ($-$CEBM Weighting):** Applying uniform evidence weighting ($w(e_i) = 1.0$) reduces Grounding to 83.7%, demonstrating that quality-stratified retrieval improves clinical synthesis.

F. Sub-Component Latency Breakdown

Table VII details the latency profile across individual stages of the Agentic Graph RAG pipeline.

G. Clinical Vignette Case Studies

Table VIII summarizes the multi-agent execution traces across four representative clinical vignettes:

1) **Case 1 — Acute Coronary Syndrome (ACS):** A 58-year-old male presenting with substernal chest pressure, diaphoresis, and ST-segment elevation in leads V2–V4. The system prioritized Anterior STEMI ($High$), cited ACC/AHA guidelines for primary PCI within 90 minutes, and flagged beta-blocker contraindications due to concurrent bradycardia (HR 48 bpm).
2) **Case 2 — Community-Acquired Pneumonia (CAP):** A 72-year-old male presenting with productive cough, fever 38.9°C, right basal crackles, and SpO2 91%. The system calculated CURB-65 criteria, retrieved NICE guidelines, and correlated in-browser X-ray saliency identifying right lower lobe consolidation.
3) **Case 3 — Severe Penicillin Allergy & Cephalosporin Clash:** A 45-year-old female with documented penicillin anaphylaxis requiring empiric antibiotic coverage for urinary tract sepsis. The Drug Intelligence Agent detected cephalosporin cross-reactivity risk, blocked amoxicillin/clavulanate, flagged cefepime for urgent clinician review, and presented guideline-supported non-beta-lactam options (ciprofloxacin / aztreonam).
4) **Case 4 — Subarachnoid Hemorrhage (SAH):** A 42-year-old female presenting with a sudden onset severe headache ("worst of life"). The system issued an urgent emergency triage alert recommending immediate non-contrast head CT (where sensitivity exceeds 98% within the first 6 hours of onset) and cautioned against NSAIDs / antiplatelets pending exclusion of intracranial hemorrhage.

TABLE IV
EMPIRICAL BENCHMARK COMPARISON ACROSS 50 MULTI-SYSTEM CLINICAL VIGNETTES (MEAN ± STD. DEV., [95% CONFIDENCE INTERVAL])

Architecture	Grounding (GR)	Top-1 DA	Top-3 DA	Safety (SC)	Traceability	Latency	Ghost†	Primary Failure Mode
Vanilla LLM	19.8% ± 6.09 [18.1, 21.5]	72.0% [59.5, 84.5]	86.0% [76.4, 95.6]	34.6% ± 8.11 [32.4, 36.9]	14.3% ± 4.13 [13.2, 15.5]	2.13 s ± 0.30	78.4%	Hallucinated PMIDs; missed drug allergies.
Vector RAG	64.8% ± 4.74 [63.5, 66.1]	76.0% [64.2, 87.8]	90.0% [81.7, 98.3]	61.3% ± 6.60 [59.5, 63.1]	56.4% ± 7.31 [54.3, 58.4]	6.48 s ± 0.80	21.2%	Misses multi-hop drug-disease contraindications.
Graph RAG	78.4% ± 3.69 [77.4, 79.5]	82.0% [71.4, 92.6]	92.0% [84.5, 99.5]	72.7% ± 5.34 [71.2, 74.2]	74.2% ± 4.40 [72.9, 75.4]	8.13 s ± 1.19	11.5%	Lacks post-hoc safety auditing and citation pruning.
MEDGUIDE (Ours)	**90.3% ± 3.94 [89.2, 91.4]**	**88.0% [79.0, 97.0]**	**96.0% [90.6, 100]**	**92.2% ± 4.15 [91.1, 93.4]**	**95.2% ± 2.76 [94.5, 96.0]**	14.68 s ± 1.65	**0.0%***	Fully grounded; safety auditor strips unverified claims.

†Ghost citation rate reported at the citation level (unverified citations / total citations generated). *0.0% ghost citations on the benchmark is achieved by deterministic design, as the Safety Auditor programmatically removes any citation identifier not present in the retrieved evidence repository $\mathcal{E}$ (0/382 citations, 95% Clopper-Pearson CI: $[0.0\%, 0.96\%]$; 0/50 vignettes, 95% CI: $[0.0\%, 7.1\%]$).

TABLE V
DOMAIN-WISE PERFORMANCE BREAKDOWN OF AGENTIC GRAPH RAG ($n = 50$)

Specialty Domain	n	Grounding	Safety	Traceability
Cardiology	12	88.7% ± 3.5	90.6% ± 4.3	94.6% ± 3.2
Pulmonology	12	91.5% ± 4.5	93.7% ± 4.8	95.9% ± 3.0
Neurology	10	90.7% ± 3.7	94.9% ± 3.7	94.9% ± 2.2
Nephrology / Endo.	8	91.7% ± 4.9	90.9% ± 2.0	95.3% ± 2.9
Infectious / Pharm.	8	89.1% ± 2.3	90.4% ± 2.9	95.5% ± 2.6
Overall Benchmark	**50**	**90.3% ± 3.9**	**92.2% ± 4.2**	**95.2% ± 2.8**

VIII. RESULTS & DISCUSSION

A. Why Agentic Graph RAG Improves Factual Grounding

The observed improvement in evidence grounding from 19.8% (Vanilla LLM) to 90.3% (Agentic Graph RAG) confirms Hypothesis **H1**. Whereas parametric models hallucinate citations when generating medical reasoning, the multi-stage pipeline decomposes intake into search query formulation, authoritative literature retrieval, and claim-level verification. Furthermore, our ablation experiments demonstrate that knowledge graph traversal and post-hoc citation auditing contribute additively: the knowledge graph preserves multi-hop relational dependencies between comorbidities, while the Safety Auditor prunes unverified assertions prior to report generation.

B. Answering RQ1: Multi-Agent Decomposition and Role Isolation

Addressing **RQ1**, multi-agent decomposition partitions the complex clinical intake task into specialized, verifiable subtasks governed by a deterministic state transition system ($\mathcal{S}$). This architecture prevents the prompt-collapse phenomenon observed in monolithic LLMs, where the model must simultaneously perform information retrieval, ontological reasoning, pharmacology checking, and textual synthesis. In Ablation 4, collapsing the 12-agent pipeline into a monolithic prompt with identical Graph RAG context caused Grounding to drop by 12.7 percentage points (90.3% $\rightarrow$ 77.6%) and Safety Compliance to plummet by 21.0 percentage points (92.2% $\rightarrow$ 71.2%). Decoupling planning, retrieval, reasoning, and auditing into discrete execution states ensures that intermediate hypothesis representations remain fully auditable and independently verifiable.

C. Deterministic Safety Verification vs Generative Variability

A key finding of this research (Hypothesis **H2**) is the necessity of decoupling safety-critical rules from stochastic LLM reasoning. In Ablation 3 ($-$Safety), removing deterministic RxNav and openFDA checks caused safety compliance to drop by 23.9 percentage points. Generative models can overlook specific drug-allergy contraindications when processing verbose clinical narratives. By encoding pharmacology rules deterministically across 7 codified high-risk medication classes, MEDGUIDE evaluates documented drug-drug and drug-allergy clashes independently of LLM prompt wording. Across the 50 benchmark vignettes, zero false alerts were observed, though parsing atypical allergen phrasing in messy real-world clinical text remains an operational boundary.

D. Answering RQ4: Multi-Signal Confidence Bands vs Diagnostic Correctness

Addressing **RQ4**, we evaluated the heuristic multi-signal composite Evidence Confidence Index (C) and its resulting ordinal bands ($\text{Band}(C) \in \{\text{High}, \text{Medium}, \text{Low}, \text{Insufficient}\}$) across the 50 benchmark vignettes against empirical diagnostic correctness:

- **High Confidence** ($C > 0.62$, $n = 38$, **76% of cases):** Exhibited a 94.7% primary diagnostic concordance (DA_1) and 97.4% safety compliance (SC), corresponding to strong multi-source guideline alignment.
- **Medium Confidence** ($0.38 < C \leq 0.62$, $n = 9$, **18% of cases):** Exhibited a 77.8% diagnostic concordance and 88.9% safety compliance, reflecting presentations with partial evidence coverage or atypical symptom overlap.
- **Low or Insufficient Confidence** ($C \leq 0.38$ **or** $|\mathcal{E}| < 3$, $n = 3$, **6% of cases):** Appropriately alerted the clinician to sparse evidentiary backing (33.3% concordance),

TABLE VI
COMPONENT ABLATION STUDY ACROSS 50 MULTI-SYSTEM CLINICAL VIGNETTES (MEAN ± STD. DEV.)

System Configuration	Grounding (GR)	Safety (SC)	Traceability (ET)	Ghost Cit.	Latency (L)
Full MEDGUIDE Framework	**90.3% ± 3.94**	**92.2% ± 4.15**	**95.2% ± 2.76**	**0.0%**	14.68 s ± 1.65
w/o Knowledge Graph (−KG)	81.2% ± 4.30	79.4% ± 5.10	82.6% ± 4.12	4.2%	14.48 s ± 1.50
w/o Safety Auditor (−Auditor)	76.5% ± 5.12	84.1% ± 4.80	81.0% ± 4.65	8.6%	12.58 s ± 1.35
w/o Pharmacovigilance Engine (−Safety)	89.1% ± 3.80	68.3% ± 6.25	88.4% ± 3.90	0.0%	12.88 s ± 1.40
w/o Agent Decomposition (Monolithic)	77.6% ± 4.10	71.2% ± 5.40	72.8% ± 4.50	9.8%	7.94 s ± 1.10
w/o CEBM Evidence Weighting (Uniform)	83.7% ± 4.55	88.5% ± 4.40	89.1% ± 3.50	0.0%	14.50 s ± 1.60

TABLE VII
SUB-COMPONENT LATENCY BREAKDOWN IN AGENTIC GRAPH RAG

Pipeline Stage / Component	Latency (s)	% Total
1. Planner Agent (LLM JSON)	1.38	9.4%
2. Biomedical Retrieval (PubMed/EPMC)	2.38	16.2%
3. Knowledge Graph Subgraph Extraction	0.20	1.4%
4. Drug & Allergy Checking (RxNav/FDA)	1.80	12.3%
5. Clinical Reasoning Agent (LLM CoT)	4.18	28.5%
6. Safety Auditor Agent (Citation Audit)	2.10	14.3%
7. Explainability & Confidence Evaluation	0.84	5.7%
8. Report Generator Agent (Markdown)	1.80	12.2%
Total End-to-End Execution	**14.68 s**	**100.0%**

prompting recommendations for further diagnostic workup rather than false reassurance.

These empirical distributions confirm that while C is an uncalibrated heuristic rather than a true Bayesian posterior probability, its ordinal bands provide meaningful risk stratification for clinical users.

E. Client-Side Radiographic Explanations and Privacy Boundary

Evaluating Hypothesis **H3**, our in-browser WebAssembly pipeline demonstrates the feasibility of computing class-specific perturbation heatmaps $S_c(g_x, g_y)$ locally without cloud image transmission (Path A). This addresses privacy constraints in multi-tenant environments. However, consistent with Arun et al. [29] and CLAIM 2024 [41], we explicitly frame saliency overlays as localization aids that direct physician attention to influential pixel patches rather than autonomous diagnostic determinants.

Furthermore, we clarify the exact operational scope of the privacy boundary: zero-egress data containment applies strictly to the client-side WebAssembly saliency localization pipeline (Path A), ensuring raw pixel arrays do not leave the clinician's browser for edge heatmaps. However, clinical text is transmitted to Google Gemini for language orchestration, longitudinal charts are stored in Google Cloud Firestore, and optional remote radiograph interpretation (Path B) utilizes Cloudinary edge storage. In production healthcare deployments, Business Associate Agreements (BAAs) and Data Protection Agreements (DPAs) with enterprise cloud providers are legally mandatory under HIPAA and GDPR.

F. Latency vs Diagnostic Verification Trade-offs

The 14.68-second execution latency of MEDGUIDE reflects an engineering trade-off:

$$\Delta L = L_{\text{MEDGUIDE}} - L_{\text{Vector RAG}} = 14.68\,\text{s} - 6.48\,\text{s} = 8.20\,\text{s} \tag{18}$$

In exchange for 8.2 seconds of additional processing, the system achieves a 25.5 percentage-point gain in evidence grounding and eliminates benchmark ghost citations. For urgent triage situations, isolated sub-agents such as Drug Intelligence and Edge Saliency execute in under 2 seconds.

IX. LIMITATIONS, SAFETY, ETHICS & MODEL TRANSPARENCY

A. Model Facts Specification

Following the clinical machine learning transparency framework established by Sendak et al. [6] and TRIPOD-LLM / CLAIM 2024 principles [41], Table IX outlines the formal Model Facts label for MEDGUIDE.

B. Ethical Governance and Alignment with WHO Guidance

The system is designed in accordance with governance principles described in the World Health Organization (WHO) 2024 guidance on Large Multi-Modal Models in health [17]:

- **Human-in-the-Loop Oversight:** Non-autonomous decision boundaries; explicit disclaimers on all generated outputs.
- **Provenance Transparency:** Direct visibility into supporting versus contradicting evidence.
- **Data Privacy:** Edge computing for radiographs and multi-tenant Firestore security rules.

C. Data Privacy vs Regulatory Scope

The client-side architecture is designed to contain raw radiographic pixels on the clinician's workstation during saliency localization (Path A); this architectural design should not be interpreted as establishing formal regulatory medical device clearance.

TABLE VIII
END-TO-END MULTI-AGENT CLINICAL VIGNETTE CASE STUDIES AND SYSTEM EXECUTION TRACES

Clinical Scenario	Patient Presentation	Prioritized Differentials	Grounded Citations	Safety Audit & Pharmacovigilance Flags
Case 1: ACS / Anterior STEMI	58M, severe retrosternal chest pressure, diaphoresis; ECG: ST-elevation V2–V4; HR 48, BP 90/60.	1. Anterior STEMI (*High*) 2. Aortic Dissection (*Moderate*) 3. Acute Myocarditis (*Low*)	ACC/AHA Revascularization (PMID: 34895950) ESC ACS Guidelines (PMID: 37622654)	**CRITICAL FLAG:** Beta-blockers (Metoprolol) contraindicated due to severe bradycardia (HR 48 bpm). Urgent Cath Lab transfer.
Case 2: Severe Pneumonia (CAP)	72M, fever 38.9°C, productive cough, SpO2 91%, CURB-65 = 3; CXR: dense RLL consolidation.	1. Community-Acquired Pneumonia (*High*) 2. Acute Pulmonary Embolism (*Mod*) 3. Heart Failure Exacerbation (*Low*)	NICE NG138 Guidelines IDSA/ATS CAP (PMID: 31573350) In-browser DenseNet Saliency	**IMAGING AID:** Saliency highlights dense right lower lobe patch. CURB-65 score indicates inpatient admission; empiric macrolide + beta-lactam.
Case 3: Penicillin Anaphylaxis	45F, dysuria, flank pain, fever 38.5°C; Documented history of Penicillin anaphylaxis (IgE-mediated).	1. Acute Pyelonephritis (*High*) 2. Complicated UTI (*High*) 3. Renal Abscess (*Low*)	IDSA Pyelonephritis (PMID: 21292654) openFDA Monograph: Ciprofloxacin	**SAFETY BLOCK:** Amoxicillin/Clavulanate blocked; Cefepime flagged for clinician review due to cross-reactivity risk. Guideline alternatives (Ciprofloxacin / Aztreonam) presented.
Case 4: Thunderclap Headache	42F, sudden onset maximal intensity headache ("worst of life"), photophobia, neck stiffness, BP 160/95.	1. Subarachnoid Hemorrhage (*High*) 2. Bacterial Meningitis (*Moderate*) 3. Cerebral Venous Thrombosis (*Low*)	AHA/ASA SAH Guidelines (PMID: 37212051) Oxford CEBM Tier 1 Review	**EMERGENCY TRIAGE:** Immediate emergent non-contrast head CT (sensitivity $> 98\%$ within 6 hours of onset, followed by LP if CT negative). Absolute contraindication to NSAIDs/antiplatelets.

TABLE IX
MEDGUIDE MODEL FACTS LABEL SPECIFICATION

Dimension	Specification / Operational Boundary
Intended Use	Evidence-based Clinical Decision Support System (CDSS) for differential diagnosis exploration.
Autonomous Diagnosis	**Strictly Prohibited.** The system operates as a decision aid and does not issue binding diagnoses.
Target Users	Licensed physicians, resident physicians, and supervised medical students.
Evidence Sources	PubMed (NCBI), Europe PMC, ClinicalTrials.gov, WHO Guidelines, openFDA, RxNorm.
Imaging & Storage	In-browser DenseNet-121 WASM saliency (Path A); logically partitioned Cloudinary media edge (`medguide-clinical/*`) with local data-URI fallback.
Chart & Provenance	7-section clinical consultation report with scannable QR verification code and structured export for standards-based EHR integration [32]–[34].
Human Oversight	**Mandatory.** Clinicians must confirm all recommendations and dosages.
Major Limitations	External biomedical API network dependencies; 50-case benchmark size; 2D single-view imaging.
Regulatory Status	Academic Research Prototype (Not FDA/CE certified as medical device).

D. Threats to Validity & Reproducibility Protocol

Following reporting best practices in TRIPOD-LLM and CLAIM 2024 [41], we categorize threats to validity:

- **Internal Validity:** Generative language model stochasticity was constrained by fixing temperature at $T = 0.2$ and top-$p = 0.95$ across all experimental runs with Gemini 2.5 Flash as the common backbone.
- **External Validity:** The benchmark comprises 50 curated clinical vignettes. While covering five acute specialties, prospective multi-center hospital validation with diverse patient cohorts remains necessary before clinical deployment.
- **Construct Validity:** The Explanation Traceability Score (ET) measures structural compliance with evidence linking and factor balance; future work will incorporate multi-reader clinician Likert evaluations.
- **Statistical Validity:** Due to bounded percentage outcomes and paired observations, we report both paired Student's t-tests and non-parametric Wilcoxon signed-rank tests with Cohen's d_z effect sizes.

X. CONCLUSION & FUTURE WORK

This paper presented **MEDGUIDE**, an explainable, multi-agent clinical decision support research prototype combining Graph RAG, multi-source literature retrieval, deterministic pharmacovigilance, logically partitioned medical storage, and client-side radiographic saliency. Evaluation across 50 multi-system clinical vignettes demonstrated a 90.3% $\pm$ 3.94% evidence grounding rate, 88.0% Top-1 diagnostic accuracy, 92.2% $\pm$ 4.15% safety compliance, and 95.2% $\pm$ 2.76% structured explanation traceability. Through deterministic post-hoc citation filtering, no ghost citations were present in final consultation outputs (0.0%, 95% Clopper-Pearson CI: $[0.0\%, 7.1\%]$ across vignettes, $[0.0\%, 0.96\%]$ across 382 citations) with high statistical significance ($p < 10^{-14}$ on grounding). Component ablation confirmed the independent necessity of knowledge graph traversal, post-hoc citation auditing, and deterministic pharmacology rules for reliable decision support. Furthermore, the platform bridges computational reasoning with clinical operations through structured 7-section consultation charts, scannable QR verification provenance, and structured export formatted for standards-based EHR integration (SMART on FHIR R4, CDS Hooks [32]–[34]).

Future work will implement and evaluate SMART on FHIR R4 and CDS Hooks interoperability adapters in hospital sandbox environments [34], [42], enrich the knowledge graph

ontology across additional UMLS and SNOMED-CT concepts, incorporate dual-view (PA/Lateral) chest neural networks, and conduct a multi-center prospective reader study with licensed clinicians.

Declarations

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Competing Interests: The authors declare that they have no competing financial, personal, or professional interests that could influence the work reported in this paper.

Data and Code Availability: All 50 benchmark clinical vignettes, evaluation harness scripts, raw paired observations, and implementation source code are openly available at https://github.com/designnest411-stack/MEDGUIDE-AI.

Ethical Compliance: This investigation evaluated purely synthetic, anonymized clinical educational vignettes and open-access public databases; no human subject trials were performed, and no private Protected Health Information (PHI) was accessed or processed.

AI-Assisted Technologies Disclosure: During the preparation of this manuscript, AI coding tools were utilized solely for code refactoring and typographical editing under the continuous supervision and verification of the human authors.

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